higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

ELECTROTECHNICS N5

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This marking guideline consists of 6 pages.

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QUESTION 1

1.1
- Increasing the brush contact resistance
- Shifting the brushes forward in a generator and backward in a motor
- Increasing the reluctance
- Interpoles
- Compensating windings (Any 3 × 1) (3)

1.2 $\text{Amp-turns}_{ARMATURE} = ½ \times I_a/c \times Z/2p = ½ \times (648/8) \times (820/8)$✓
$= 4\,151{,}25$ At(s)✓

$4\,151{,}25/\text{Amp-turns}_{Interpole} = 0{,}88$✓
$\text{Amp-turns}_{Interpole} = 4\,151{,}251/0{,}88 = 4\,717{,}329$ At(s)✓

$\text{Turns}_{Interpole} = 4\,717{,}329/648 = 7{,}279$ turns
$= 7$ turns✓ (5)

1.3 $R_1 = V_S/I_{st} = 480/115 = 4{,}174\ \Omega$✓

$K = \sqrt[n]{R/r} = \sqrt[15]{\frac{4{,}174}{0{,}35}}$ ✓ $= 1{,}179$✓

$r\,1 = R_1[(K - 1)/K] = 4{,}174[(1{,}179 - 1)$✓$/1{,}179] = 0{,}63\ \Omega$✓

$r\,2 = 0{,}636/1{,}179 = 0{,}539\ \Omega$✓

$r\,3 = 0{,}539/1{,}179 = 0{,}458\ \Omega$✓ (7)

1.4 $\text{Amp-turn}_{NL} = 2{,}5 \times 1\,350 = 3\,375$✓
$\text{Amp-turn}_{L} = 4{,}5 \times 1\,350 = 6\,075$✓

$I_L = 120 \times 10^3/440 = 272{,}727$ A✓

$\text{Amp-turns}_{SERIES} = 6\,075 - 3\,375 = 2\,700$✓

$I_a = I_L + I_F = 272{,}727 + 2{,}5 = 275{,}227$ A

$\text{Turns}_{SERIES} = 2\,700/275{,}227 = 9{,}8$✓

Say: 10 turns (5)
[20]

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QUESTION 2

2.1 2.1.1 $F_r = (2\pi \sqrt{LC})^{-1}$
$1,4 \times 10^6 = (2\pi \sqrt{0,000185 \times C})^{-1}$✓
$\sqrt{0,000185 \times C} = (2\pi \times 1,4 \times 10^6)^{-1}$
$C = 1,292 \times 10^{-14}/185 \times 10^{-6}$✓
$= 6,986 \times 10^{-11}$
$= 69,857$ pF✓
OR

$X_L = X_C = 2\pi \times 1,4 \times 10^6 \times 185 \times 10^{-6}$
$= 1\,627,345\ \Omega$✓
$1\,627,345 = (2\pi\ 1,4 \times 10^6 \times C)^{-1}$✓
$C = (2\pi \times 1,4 \times 10^6 \times 1\,627,345)^{-1}$
$= 6,986 \times 10^{-11}$✓
$= 69,857$ pF (3)

2.1.2 $Z_D = Z_P = L/CR = 185 \times 10^{-1}/69,857 \times 10^{-12} \times 8)$✓
$= 331\,033,397\ \Omega$✓
$= 331,033\ k\Omega$ (2)

2.1.3 $Z_T = Z_P + Z_{SE} = 331\,033,397 + 6\,580 = 337\,613,397\ \Omega$✓

$I_T = V_T/Z_T = 440/337\,613,397 = 1,303 \times 10^{-3}$ A✓
$= 1,303$ mA

$V_P = I_T Z_P = 1,303 \times 10^{-1} \times 331\,033,397 = 431,425$ V✓

$Z_{COIL} = 8 + j1\,627,345 = 1\,627,365 < 89,72\ \Omega$
$I_{COIL} = V_P/Z_{COIL} = 431,425/1\,627,36 = 0,265$ A✓

$I_C = V_P/X_C = 431,425/1\,627,354 = 0,265$ A $= 265,108\ \mu$A✓ (5)

2.2 2.2.1 $X_{L1} = X_{C1} = (2\pi \times 0,95 \times 10^6 \times 287 \times 10^{-12})^{-1} = 583,734\ \Omega$✓
$L = X_{L1}/(2\pi f) = 583,734/(2\pi \times 0,95 \times 10^6)$✓
$= 97,794 \times 10^{-6}$ H $= 97,794\ \mu$H✓ (3)

2.2.2 $X_{C2} = (2\pi \times 0,95 \times 10^6 \times 271 \times 10^{-12})^{-1} = 618,197\ \Omega$✓
$X_C = X_{C2} - X_{C1} = 618,197 - 583,734 = 34,463\ \Omega$✓
$Z_2^2 = R^2 + (34,463)^2$✓
$I_2 = 0,74\ I_1 = 0,74\ V/Z_1$
$0,74\ V/Z_1 = V/Z_2$
$R^2 + (34,463)^2 = (1,351\ R)^2$✓
$R^2 + (34,463)^2 = 1,825\ R^2$
$0,825\ R^2 = (34,463)^2$✓
$R^2 = 1\,439,634$✓
$R = \sqrt{1\,439,634}$
$= 37,943\ \Omega$✓ (7)
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QUESTION 3

3.1 Iron losses will increase with an increase of frequency. (2)

3.2 When the transformer has a leading power factor (1)

3.3 $I_1 = S/V_1 = 125 \times 10^3/2\,400\checkmark = 52{,}083\ A\checkmark$

$\%R = I_1 R_{E1}/V_1$
$3/100 = 52{,}083 \times R_{E1}\checkmark/2\,400$
$R_{E1} = 1{,}382\ \Omega\checkmark$

$\%X = I_1 X_{E1}/V_1$
$0{,}042 = 52{,}083\ X_{E1}\checkmark/2\,400$
$X_{E1} = 1{,}935\ \Omega\checkmark$

$Reg = I_1(R_{E1}\ Cos\Theta + X_{E1}\ Sin\Theta)/V_1$
$= 52{,}083[(1{,}382 \times 0{,}86) + (1{,}935 \times Sin30{,}68)]\checkmark/2\,400$
$= 52{,}083 \times 2{,}176/2\,400\checkmark$
$= 0{,}0472$ p.u.✓ $= 4{,}722\%$

$V_{2FL} = V_{2NL} - Reg \times V_{2NL} = 600 - (0{,}0472 \times 600)\checkmark$
$= 600 - 28{,}32 = 571{,}68\ V\checkmark$ (11)

3.4 3.4.1 $I_2 = S/V_2 = 13 \times 10^3/200\checkmark = 65\ A\checkmark$

$\%R = I_2 R_{E2}/V_2$
$0{,}02 = 65 \times R_{E2}/200\checkmark$
$R_{E2} = 400/6\,500 = 0{,}062\ \Omega\checkmark$ (4)

3.4.2 $\%X = I_2 X_{E2}/V_2$
$0{,}04 = 65\ X_{E2}/200\checkmark$
$X_{E2} = 800/6\,500 = 0{,}123\ \Omega\checkmark$ (2)

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QUESTION 4

4.1 $P_{IN} = P_O/\eta = 20 \times 10^3/0{,}88 = 22{,}727$ kW✓
$P_{IN} = \sqrt{3}\ V_L I_L Cos\Theta$
$V_L I_L = P_{IN}/(\sqrt{3}\ Cos\Theta) = 22{,}727 \times 10^3/(\sqrt{3} \times 0{,}89)\checkmark = 14\,743{,}191\ A\checkmark$

$\Theta = Cos^{-1}(0{,}89) = 27{,}13^\circ\checkmark$
$W_1 = V_L I_L Cos(30 - \Theta) = 14\,743{,}191 \times Cos(30 - 27{,}13)\checkmark$
$= 14\,724{,}699\ W = 14{,}725$ kW✓

$W_2 = V_L I_L Cos(30 + \Theta) = 14\,724{,}699 \times Cos(30 + 27{,}13)\checkmark$
$= 8\,001{,}642\ W = 8{,}002$ kW✓ (8)

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4.2 4.2.1 $X_L = 2\pi \times 50 \times 465 \times 10^{-3} = 8\ \Omega$✓
$V_P = V_L/\sqrt{3} = 25 \times 10^3/\sqrt{3} = 14\,433{,}757$ V✓
$V_R = V_P(\cos\Theta + j\sin\Theta) = 14\,433{,}757\ (0{,}8 + j0{,}6)$✓
$= 11\,547{,}006 + j8\,660{,}254$✓

$I_P = 12 \times 10^6/(\sqrt{3} \times 25 \times 10^3 \times 0{,}8) = 346{,}41$ A✓
$V_d = I_P Z = 346{,}41(4 + j8)$
$= 1\,385{,}64 + j2\,771{,}28$✓

$V_S = V_R + V_d = 11\,547{,}006 + j8\,660{,}254 + 1\,385{,}64 + j2\,771{,}28$✓
$= 12\,932{,}646 + j11\,431{,}534$✓
$= 17\,260{,}745 < 41{,}47$ V✓ (9)

4.2.2 $Reg = (V_S - V_R)/V_R = (17\,260{,}745 - 14\,433{,}757)$✓$/14\,433{,}757$
$= 19{,}586\%$✓ (2)

4.3 $P_T = 820 + 300 = 1\,120$ W✓ (1)
[20]

QUESTION 5

5.1 A high voltage decreases both the power factor and the slip but increases the torque. (2)

5.2 There is no relative speed difference between the rotating magnetic field and the rotor.✓ No EMF will be induced in the rotor✓ and therefore there will be no rotor current✓ and no torque.✓ (4)

5.3 5.3.1 $P = 60f/N_S = (60 \times 50)/1\,500$✓ $= 2$ (pairs)
Poles $= 2 \times 2 = 4$ poles✓ (2)

5.3.2 $E_P = 500/\sqrt{3} = 288{,}675$ V✓
$Z_P = 10 \times 6 \times 4 = 240$ cond/ph✓
$\phi = E_P/(2{,}22\ K_d\ K_P\ Z_P\ f)$
$= 288{,}675/(2{,}22 \times 0{,}97 \times 240 \times 50)$✓
$= 0{,}0112$ Wb✓
$= 11{,}161$ MWb (4)

5.4 5.4.1 $N_S = 60f/p = (60 \times 50)/3 = 1\ 000$ r/min✓
$S = (N_S - N_R)/N_S$
$(0,06 \times 1\ 000) = 1\ 000 - N_R$
$N_R = 940$ r/min✓ (2)

5.4.2 $P_O = 2\pi N_R T/60$
$45\ 285 \times 60 = 2\pi \times 940 \times T$✓
$T = 460,042$ Nm✓

Rotor input $= (2\pi \times 1\ 000 \times 460,042)$✓$/60$
$= 48\ 175,486$ W✓

Rotor copper loss $= S \times$ Rotor input
$= 0,06 \times 48\ 175,486$✓
$= 2\ 890,529$ W✓ (6)
[20]

TOTAL: 100

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